

PRATEEK SINGH

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AI Engineering Intern | LLM Agents, RAG Systems, Vision AI, Async Python | Relocation Ready - Bangalore

Summary

AI Engineering Intern with hands-on experience building **AI agents, multi-agent systems, RAG pipelines, vision-AI workflows, backend AI components, and healthcare-focused GenAI products**. Proven track record of reducing operational latency by **30%**, automating document processing by **70%**, and achieving **85% emotion analysis accuracy** in a multimodal AI product. Skilled in **Python, FastAPI/Flask, async/await basics, REST APIs, Groq/Ollama, OpenAI/Anthropic/Gemini API awareness, Supabase/PostgreSQL, OpenCV, DeepFace, PyTorch, ONNX, prompt engineering, execution tracing, and vector database concepts**. Strong interest in **healthcare AI, skin-analysis vision systems, safe LLM agents, tool calling, and reliable RAG applications**.

Work Experience

Hestabit Technologies Pvt. Ltd.

Noida Sec-63, India

AI Engineering Intern – LLM & Backend AI

Jan 2026 – Present

- **Achieved 30% reduction in decision-making latency** across 4 internal AI project areas by engineering automated workflows, RAG retrieval pipelines, FastAPI-style backend modules, and structured outputs.
- **Built production-ready LLM components** for Multi-RAG, n8n automation, AI-engineered platform, and internal systems by implementing modular LLM workflows, prompt-driven automation, validation logic, and API integrations.
- **Improved AI system reliability** for asynchronous agent tasks by designing retry/fallback logic, execution tracing, response-quality checks, tool-flow documentation, and monitoring-ready debugging outputs.

NITI Aayog, Ministry of Home Affairs

New Delhi, India

Python Automation & Data Intern

Sept 2025 – Nov 2025

- **Reduced manual data preparation effort by 50%** across multi-departmental datasets by developing Python extraction, cleaning, normalization, transformation, validation, and reporting workflows.
- **Enhanced data transparency** for senior stakeholders by converting raw policy datasets into structured reports, validation rules, Power BI/Excel outputs, and evidence-ready insights.
- **Improved repeatability of analytical workflows** across S&T indicators by documenting processing logic, quality checks, transformation rules, and handoff-ready outputs.

AlgoStrauss IT Services and Consulting LLP

Remote

Database Engineer Intern – SQL & Data Systems

Sept 2024 – Nov 2024

- **Improved query response time by 20%** across **100K+ records** by refactoring nested subqueries, optimizing joins, applying indexing strategies, and validating query outputs.
- **Strengthened backend data reliability** for reporting workflows by building reusable SQL logic, consistency checks, query verification steps, and maintainable database handoff notes.
- **Supported AI-ready data handling** by cleaning structured records, validating transformations, improving consistency, and preparing database-backed analytical outputs.

Ghadai Technology

Gurugram, Haryana

Data Analyst Intern – Healthcare Analytics

Jul 2024 – Sept 2024

- **Improved healthcare data integrity by 25%** across **5+ datasets** by building Python-based data cleaning, validation, anomaly detection, and reporting workflows.
- **Prepared analysis-ready healthcare datasets** for business users by applying preprocessing, trend analysis, KPI reporting, validation checks, and dashboard workflows.
- **Enabled faster healthcare review** for analytics teams by converting raw records into anomaly summaries, trend views, and stakeholder-ready insights.

Projects

NeuroSense AI – Multimodal Healthcare AI Platform | Flask, Supabase, Groq, Ollama, OpenCV, DeepFace | [GitHub](#)

- **Situation/Task:** Traditional wellness tools lacked emotional context, vision signals, session history, and safe response behavior for sensitive user interactions.
- **Action:** Developed Flask/Supabase backend, integrated OpenCV/DeepFace emotion analysis with Groq/Ollama LLM workflows, and engineered safety-aware prompts with persistent memory.
- **Result:** Achieved **85% emotion analysis accuracy** and delivered a context-aware AI interaction engine relevant to vision-based healthcare and skin-analysis workflows.

n8n Multi-Agent Workflow & Orchestration Dashboard | *n8n, Next.js, PostgreSQL, Supabase, Ollama*

- **Situation/Task:** Multi-step AI agent executions were hard to monitor because tool-flow failures, routing decisions, latency, and bottlenecks were hidden.
- **Action:** Built n8n workflows with execution tracing, structured logs, PostgreSQL-backed session memory, retry/fallback logic, scorecards, and webhook routing.
- **Result:** Made AI-agent workflows traceable, reliable, and easier to debug while proving readiness for orchestration, tool-calling, and agent-state monitoring.

AI Document Automation & Knowledge Retrieval | *Python, LLMs, Prompt Engineering, PDF Extraction, JSON, Excel*

- **Situation/Task:** Manual review of PDFs and text notes created slow, inconsistent extraction for reports, summaries, and knowledge-ready workflows.
- **Action:** Built an LLM-based RAG-style workflow with reusable prompts, structured schemas, fallback prompts, validation checks, and JSON/Excel output formatting.
- **Result:** Reduced document processing time by **70%** and improved repeatability through documented extraction rules, output schemas, and review-ready summaries.

Scrap Simulation – Vision AI Classification Pipeline | *Python, PyTorch, ResNet18, ONNX, OpenCV, Docker* | [GitHub](#)

- **Situation/Task:** Real-time visual classification required accurate model predictions, clean preprocessing, and low-latency inference across 5 image categories.
- **Action:** Developed a PyTorch/ResNet18 model, applied OpenCV preprocessing, evaluated outputs, optimized inference using ONNX, and containerized execution.
- **Result:** Achieved **94% classification accuracy** and reduced inference latency by **25%**, demonstrating full-cycle vision AI engineering.

Research

Multimodal Agentic AI Platform for Mental Wellness Analytics | *Under Review / Ongoing Research*

- Research explores agentic AI workflows, multimodal wellness analytics, emotion-aware user state, context persistence, responsible AI support, and safety-aware response orchestration.
- Connected research implementation with NeuroSense AI by studying how LLM agents can combine text, vision-based emotion signals, memory, and workflow safeguards for health-tech use cases.

Leadership

Vice President, AILYTICS Club | *AI/ML Student Community*

- Led technical learning initiatives for **100+ students** on AI/ML project building, prompt engineering, backend AI workflows, documentation, and practical GenAI use cases.

Education

SRM University, Delhi-NCR

B.Tech – Computer Science & Engineering | Specialization: Data Science & AI

2022 – 2026

CGPA: 8.5

Skills & Certifications

AI Engineering: AI Agents, Multi-Agent Systems, Agent Orchestration, Tool Calling, LLM Applications, Prompt Engineering, AI Evaluation, Response Quality Checks

LLM APIs & RAG: OpenAI API Awareness, Anthropic API Awareness, Gemini API Awareness, Groq, Ollama, RAG Pipelines, Embeddings, Pinecone/FAISS/ChromaDB Concepts

Vision AI & ML: Computer Vision, OpenCV, DeepFace, PyTorch, ONNX, Image Preprocessing, Model Evaluation, Inference Optimization, Skin-Analysis Interest

Backend & Async: Python, Async/Await Basics, FastAPI, Flask, REST APIs, Webhooks, API Integration, Supabase, PostgreSQL, MySQL, SQL

Tools & Practices: Docker, Git/GitHub, Postman, Jupyter Notebook, Linux Basics, Debugging, Technical Documentation, CI/CD Basics

Healthcare AI Fit: Healthcare Analytics, Sensitive AI Workflows, Safety-Aware Response Design, User Context, Responsible AI, Vision-Based Analysis

Leadership & Coursework: Vice President - AILYTICS Club, Led AI/ML Workshops for 100+ Students, DSA, DBMS, ML, AI, Software Engineering

Certifications: Artificial Intelligence Analyst – IBM | OCI Data Science Professional | Technovate 2.0 – IBM | Machine Learning Using R | Introduction to Big Data, Hadoop and the Ecosystems